

```

1  #INCLUDE <STDIO.H>
2  #INCLUDE <STRING.H>
3  #INCLUDE <CTYPE.H>
4
5  INT MAIN() {
6      CHAR STR[500];
7      CHAR WORDS[100][50];
8      INT FREQ[100];
9      INT WORDCOUNT = 0;
10
11     PRINTF("ENTER A STRING: ");
12     GETS(STR);
13
14     // STEP 1: SPLIT STRING INTO WORDS
15     CHAR *TOKEN = STRTOK(STR, " ,.-!?");
16     WHILE (TOKEN != NULL) {
17         STRCPY(WORDS[WORDCOUNT], TOKEN);
18         FREQ[WORDCOUNT] = 1;
19         WORDCOUNT++;
20         TOKEN = STRTOK(NULL, " ,.-!?");
21     }
22
23     // STEP 2: COUNT FREQUENCY OF EACH WORD
24     FOR (INT I = 0; I < WORDCOUNT; I++) {
25         FOR (INT J = I + 1; J < WORDCOUNT; J++) {
26             IF (STRCMP(WORDS[I], WORDS[J]) == 0) {
27                 FREQ[I]++;
28                 FREQ[J] = -1; // MARK AS COUNTED
29             }
30         }
31     }
32
33     // STEP 3: FIND MAX FREQUENCY
34     INT MAXFREQ = 0;
35     INT MAXINDEX = 0;
36
37     FOR (INT I = 0; I < WORDCOUNT; I++) {
38         IF (FREQ[I] > MAXFREQ) {
39             MAXFREQ = FREQ[I];
40             MAXINDEX = I;
41         }
42     }
43
44     // STEP 4: OUTPUT RESULT
45     PRINTF("\nWORD WITH MAXIMUM FREQUENCY:\n");
46     PRINTF("%s" APPEARS %d TIMES\n", WORDS[MAXINDEX], MAXFREQ);
47
48     RETURN 0;
49 }
50

```